



# THEORETICAL FRAMEWORK

The study is grounded on four theories:



## OPTIMIZATION & SCHEDULING THEORY

- **Diet Problem Foundations:** Converts nutritional and logistical requirements into deterministic mathematical constraints and objective functions.
- **Slot-Based Assignment:** Treats weekly planning as a scheduling problem across a fixed horizon of 7 days and 21 meal slots.
- **Objective Trade-offs:** Automatically balances competing goals, such as keeping calories low while maximizing meal variety and budget rules.



## ALGORITHMIC MODEL (MILP & CP-SAT)

- **Binary Decision Variables:** Formulates selection criteria using binary recipe-slot assignment variables where candidate recipes are encoded as 0 or 1.
- **Two-Stage Algorithmic Approach:**
  - **Stage 1 (Preference-Aware Candidate Construction):** Executes a deterministic filtering layer, heuristic scoring, and similarity pruning.
  - **Stage 2 (CP-SAT Assignment):** Uses Google OR-Tools to solve the final weekly schedule based on constraints.
- **State-Space Reduction:** Reduces the candidate search space in the first stage to lessen the solver burden, ensuring the system generates schedules within acceptable execution times.



## RULE-BASED SYSTEMS & ASSISTIVE ML

- **Deterministic Guardrails:** Implements strict logical conditions to handle unbreakable safety rules like allergen blocks and budget limits.
- **LightGBM Heuristic Ranker:** Uses a tree-based machine learning model as a helper to score and prioritize culturally familiar Filipino meals.
- **Non-Authoritative Component:** Assistive machine learning does not override medical hard rules or final optimization constraints.

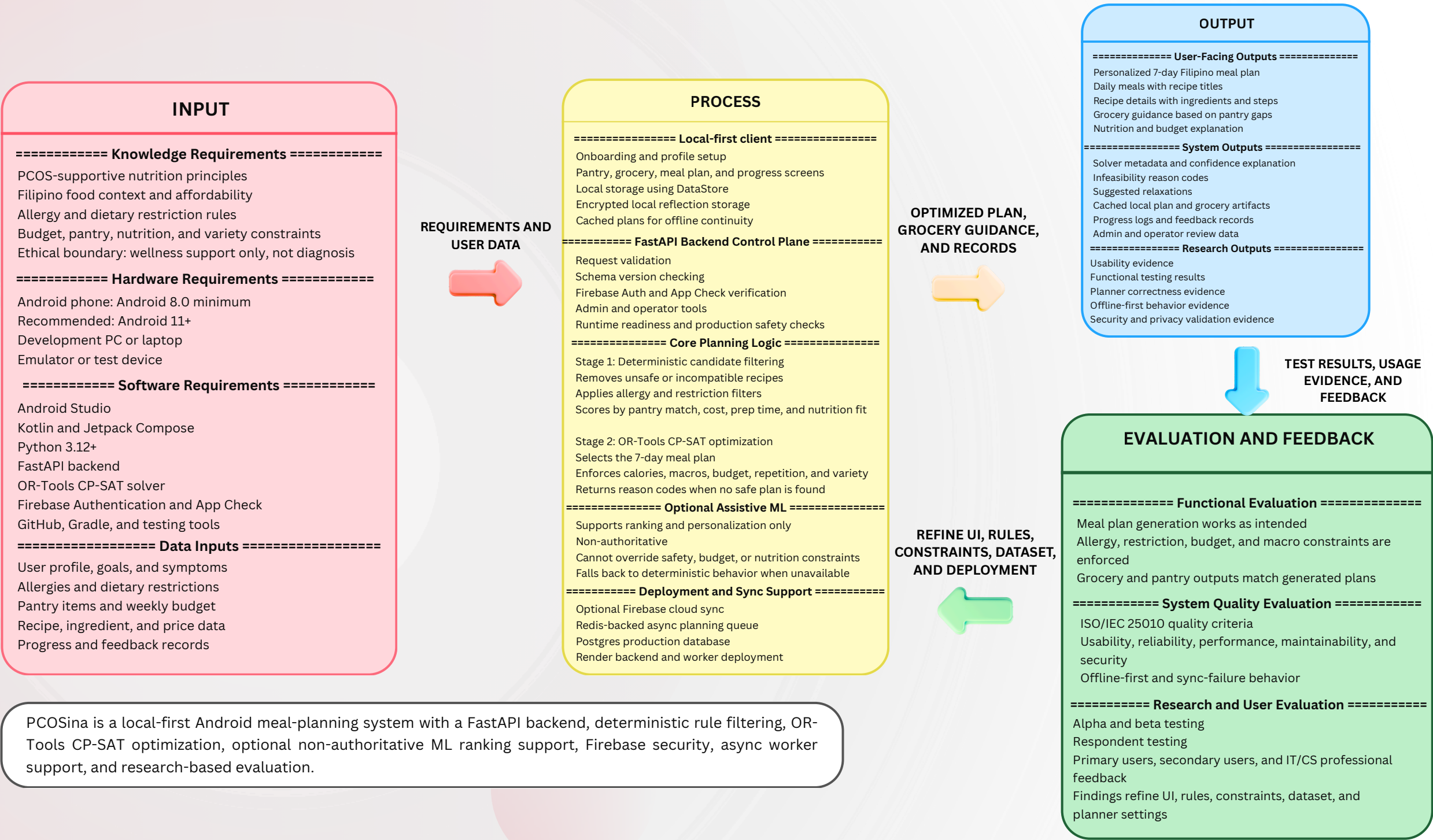


## RULE-BASED SYSTEMS & ASSISTIVE ML

- **Offline-First Architecture:** Preserves continuous user access to saved profiles, meal tables, and grocery records using DataStore, encrypted shared preferences, and local artifacts.
- **Modular Architecture:** Separates independent modules for better system testing and traceability.
- **Privacy-Conscious Data Handling:** Restricts server interaction exclusively to optimization inputs, logs and tracking stay securely on-device.



# CONCEPTUAL FRAMEWORK





# ALGORITHM

## Two-Stage MILP Meal Planning Algorithm (Flowchart)



### CORE IDEA

- Solver-first hybrid meal-planning algorithm
- Deterministic filtering + guarded ML ranking + CP-SAT
- Designed for feasibility, safety

### STAGE 1: RULE-BASED FILTERING

- Prunes infeasible recipes using restrictions and exclusions
- Prioritizes pantry matches and budget feasibility
- Caps candidate set to reduce search space and solver load

### STAGE 2: MILP OPTIMIZATION

- Binary decision variables for weekly meal selection
- Enforces nutrition, budget, and variety constraints
- Objective balances nutrition alignment, variety, and practical cost

### ROBUSTNESS & PERSONALIZATION

- Fallback greedy strategy when constraints are too strict
- Supports preferences, calories, macros, budget, and pantry use
- Advanced attributes reserved as future constraint extensions



# QUADRANT

Theoretical, Conceptual  
Framework and Algorithm

Statement of the  
Problem and Objectives

System  
Architecture

Compliance  
Report

Chapter 4

Chapter 5

Chapter 6

## SYSTEM ARCHITECTURE

### Zone E: Output and Improvement

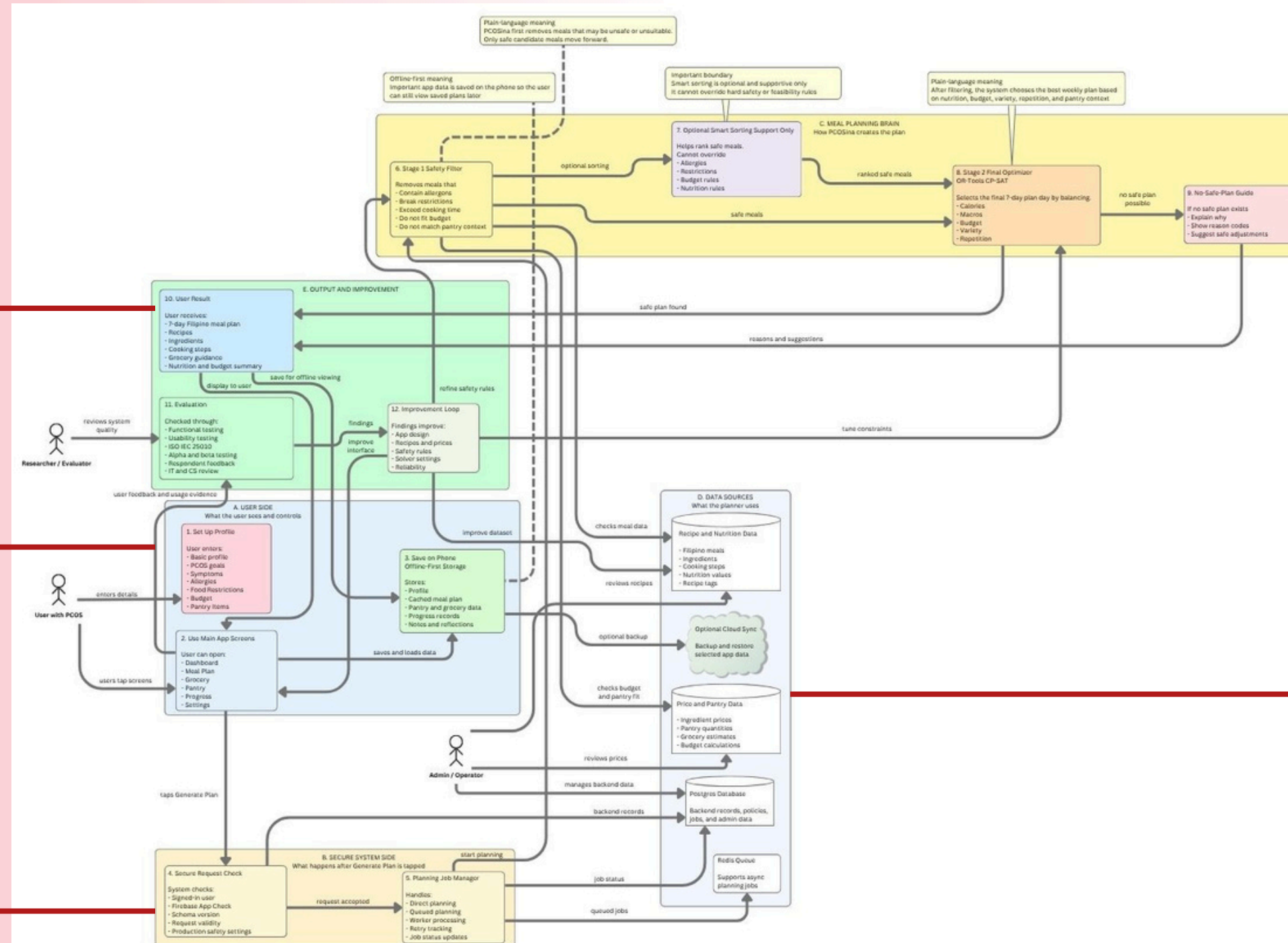
- Tracks backend processing metrics and algorithmic code correctness via ISO/IEC 25010 software quality characteristics conducted by CS/IT experts.
- Feeds verification vectors back into system data layers combining clinical boundary checks from RNDs and OB-GYNs with usability reviews from primary users managing PCOS.
- Continuously hardens, refines, and tunes constraints to maintain strict non-diagnostic limits.

### Zone A: User Side

- Manages user onboarding, profile settings, dynamic symptom tracking, pantry inventory, and localized grocery lists.
- Follows an offline-first design using Android DataStore. Users can access previously generated meal plans, ingredients, and tracking data directly on device storage without an active internet connection.

### Zone B: Secure System Side

- Intercepts out-of-device system routing utilizing Firebase App Check verification and incoming request schema validation.
- Queues high-computation job workers through an asynchronous Redis-backed message architecture.
- Acts as the authoritative gatekeeper to protect backend resources from bad requests before processing payload arrays.



### ZONE C: Meal Planning Brain

- Stage 1 (Preference-Aware Filtering):** Processes raw recipe collections through deterministic hard safety constraints, allergy exclusion parameters, and basic budget boundaries.
- Stage 2 (CP-SAT Assignment):** Converts the filtered collection into a MILP-style 0-1 logic matrix, utilizing the Google OR-Tools CP-SAT framework across a 21-meal slot grid.
- Guarantees structural optimization targets while ensuring the system generates schedules within acceptable execution times.

### ZONE D: Data Sources

- Hosts localized price references, ingredient tokens, and curated Filipino-context meal database tables within a PostgreSQL engine.
- Handles supplementary cloud synchronization routines separately to ensure system processes remain non-blocking.
- Serves as the immutable source of truth for the Stage 1 preference construction layer.